

Tutorial – 02

1. Predict and write output for the following code.

```
using System;
namespace DecisionMaking
{
    class Program
    {
        static void Main(string[] args)
        {
            /* local variable definition */
            int a = 10;
            /* check the boolean condition using if statement */
            if (a < 20) {
                /* if condition is true then print the following */
                Console.WriteLine("a is less than 20");
            }
            Console.WriteLine("value of a is : {0}", a);
            Console.ReadLine();
        }
    }
}
```

Code:

```
using System;

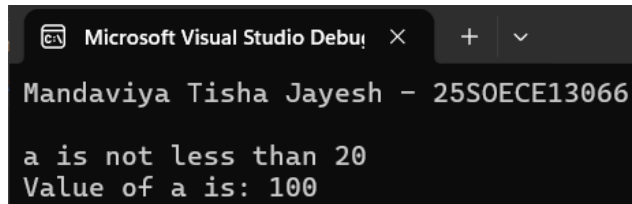
namespace Practicals
{
    internal class T2_01
    {
        public static void T2_01Main()
        {
            int a = 100;
            if (a < 20) {
                Console.WriteLine("a is less than 20");
            }
            else {
                Console.WriteLine("a is not less than 20");
            }
        }
    }
}
```

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```
        Console.WriteLine("value of a is: " + a);
    }
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
a is not less than 20
Value of a is: 100
```

2. Write missing statement to get the desired output.

```
using System;
namespace DecisionMaking
{
    class Program
    {
        static void Main(string[] args)
        {
            /* local variable definition */
            int a = 100;
            /* check the boolean condition */
            if (a < 20) {
                /* if condition is true then print the following */
                Console.WriteLine("a is less than 20");
            }
            else {
                /* if condition is false then print the following */
                //.....Missing statement-1.....//
            }
            //.....Missing statement-2.....//
            Console.ReadLine();
        }
    }
}
```

Output

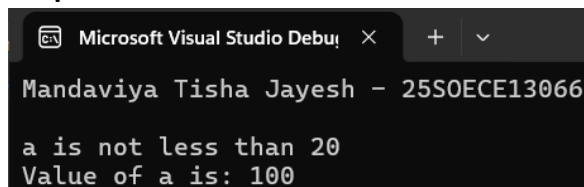
```
a is not less than 20
value of a is : 100
```

Code:

```
using System;

namespace Practicals
{
    internal class T2_02
    {
        public static void T2_02Main()
        {
            int a = 100;
            if (a < 20)
            {
                Console.WriteLine("a is less than 20");
            }
            else
            {
                Console.WriteLine("a is not less than 20");
            }
            Console.WriteLine("value of a is: " + a);
        }
    }
}
```

Output:



Microsoft Visual Studio Debug Console window showing the output of the program. The window title is 'Microsoft Visual Studio Debug Console'. The output text is: 'Mandaviya Tisha Jayesh - 25SOECE13066', 'a is not less than 20', and 'Value of a is: 100'.

3. Correct the following code and write output for the corrected code.

```
using System;
namespace ConsoleApplication1 {
    class Program {
        static void Main(string[] args) {
            char firstName = "John";
            char lastName = "Doe";
            Console.WriteLine("Name: " + firstName + " " + lastName);
            Console.WriteLine("Please enter a new first name:")
            firstName = Console.ReadLine();
            Console.WriteLine("New name: " + firstName + " " + lastName);
        }
    }
}
```

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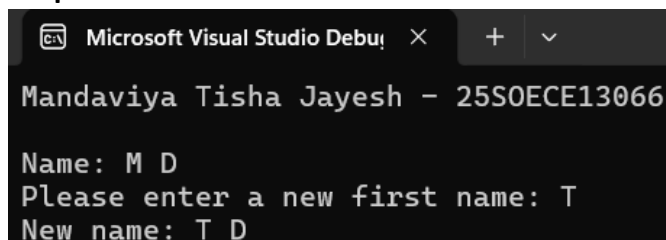
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```
        Console.ReadLine();  
    }  
}  
}
```

Code:

```
using System;  
  
namespace Practicals  
{  
    internal class T2_03  
    {  
        public static void T2_03Main()  
        {  
            char firstName = 'J';  
            char lastName = 'D';  
            Console.WriteLine("Name: " + firstName + " " + lastName);  
  
            Console.WriteLine("Please enter a new first name:");  
            firstName = Convert.ToChar(Console.ReadLine());  
  
            Console.WriteLine("New name: " + firstName + " " + lastName);  
        }  
    }  
}
```

Output:



Microsoft Visual Studio Debug Console window showing the output of the program. The title bar reads 'Microsoft Visual Studio Debug Console'. The output text is: 'Mandaviya Tisha Jayesh - 25SOECE13066', 'Name: M D', 'Please enter a new first name: T', and 'New name: T D'.

4. Input two number A and B. perform different operations using different operators and different data types available in C#. (Note : Follow all the operators and data types to do above task. Use Online help whenever necessary.)

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T2_04
```

```
    {
```

```
        public static void T2_04Main()
```

```
        {
```

```
            Console.WriteLine("Enter a:");
```

```
            int a = Convert.ToInt32(Console.ReadLine());
```

```
            Console.WriteLine("Enter b:");
```

```
            int b = Convert.ToInt32(Console.ReadLine());
```

```
            Console.WriteLine("\n---Arithmetic Operator:---");
```

```
            Console.WriteLine("Addition is: " + (a + b));
```

```
            Console.WriteLine("Subtraction is: " + (a - b));
```

```
            Console.WriteLine("Multiplication is: " + (a * b));
```

```
            Console.WriteLine("Division is: " + (a / b));
```

```
            Console.WriteLine("\n---Relational Operator:---");
```

```
            Console.WriteLine("a == b: " + (a == b));
```

```
            Console.WriteLine("a != b: " + (a != b));
```

```
            Console.WriteLine("a > b: " + (a > b));
```

```
            Console.WriteLine("a < b: " + (a < b));
```

```
            Console.WriteLine("\n---Logical Operator:---");
```

```
            Console.WriteLine("a > b && b > a: " + (a > b && b > a));
```

```
            Console.WriteLine("a > b || b > a: " + (a > b || b > a));
```

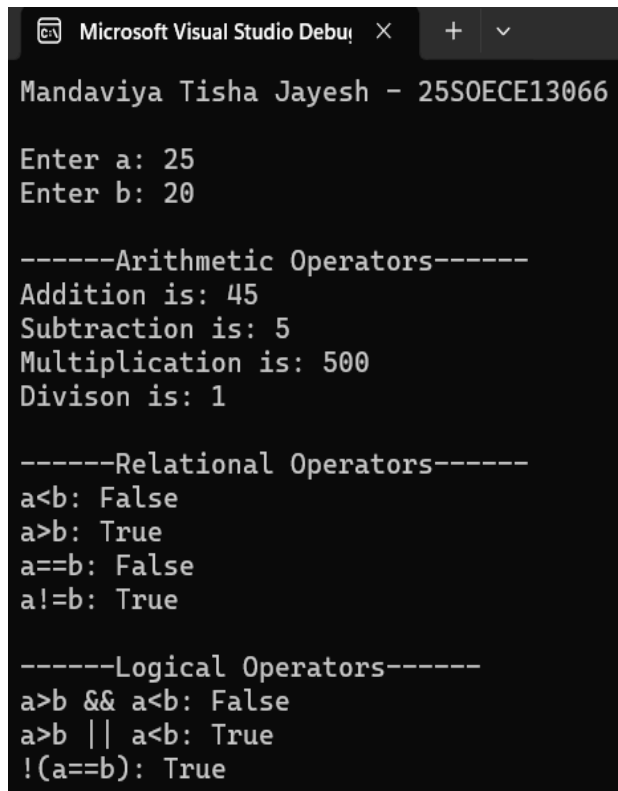
```
            Console.WriteLine("!(a == b): " + (!(a == b)));
```

```
        }
```

```
    }
```

```
}
```

Output:



```
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Enter a: 25
Enter b: 20

-----Arithmetic Operators-----
Addition is: 45
Subtraction is: 5
Multiplication is: 500
Divison is: 1

-----Relational Operators-----
a<b: False
a>b: True
a==b: False
a!=b: True

-----Logical Operators-----
a>b && a<b: False
a>b || a<b: True
!(a==b): True
```

5. Rearrange the given code to correct the program. The resultant program will be to enter 5 elements into an array and print sum of these elements.

```
using System;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            for (int i = 0; i < 5; i++) {
                string str = Console.ReadLine();
            }
            for (int i = 0; i < 5; i++) {
                sum = sum + arr[i];
            }
            Console.WriteLine("Sum of Elements : {0}",sum);
            int[] arr = new int[5];
            int sum = 0;
        }
    }
}
```

```
        arr[i] = Convert.ToInt32(str);
        Console.WriteLine("Enter Element {0}: ", i);
        Console.Read();
    }
}
```

Output:

```
Enter Element 0: 1
Enter Element 1: 2
Enter Element 2: 3
Enter Element 3: 4
Enter Element 4: 5
Sum of Elements : 15
```

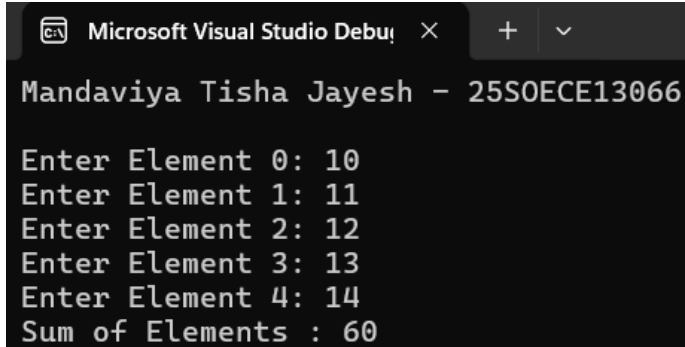
Code:

```
using System;

namespace Practicals
{
    internal class T2_05
    {
        public static void T2_5Main()
        {
            int[] arr = new int[5];
            int sum = 0;

            for (int i = 0; i < 5; i++)
            {
                Console.WriteLine("Enter Element {0}: ", i);
                string str = Console.ReadLine();
                arr[i] = Convert.ToInt32(str);
            }
            for (int i = 0; i < 5; i++)
            {
                sum = sum + arr[i];
            }
            Console.WriteLine("Sum of Elements : {0}", sum);
        }
    }
}
```

Output:



```
Microsoft Visual Studio Debug Console
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Enter Element 0: 10
Enter Element 1: 11
Enter Element 2: 12
Enter Element 3: 13
Enter Element 4: 14
Sum of Elements : 60
```

6. Write missing statement to get the desired output.

```
using System;
public class Hello3
{
    public static void Main(string[] args)
    {
        Console.WriteLine("Hello, World!");
        Console.WriteLine("You entered the following {0} command line arguments:", args.Length );
        //.....Missing statement-1.....//
        //.....Missing statement-2.....//
        //.....Missing statement-3.....//
        //.....Missing statement-4.....//
    }
}
```

Output:

```
-----
Hello, World!
You entered the following 4 command line arguments:
A
B
C
D
```

Code:

```
using System;
```

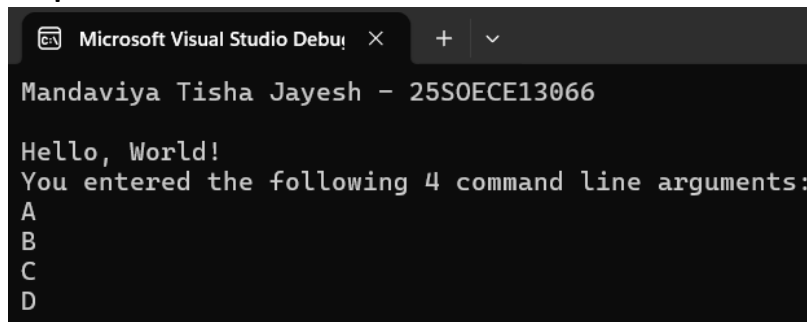
```
namespace Practicals
```


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```
{
    internal class T2_06
    {
        public static void Main(string[] args)
        {
            Console.WriteLine("Mandaviya Tisha Jayesh - 25SOECE13066\n");
            Console.WriteLine("Hello, World!");
            Console.WriteLine("You entered the following {0} command line arguments:",
args.Length);
            for (int i = 0; i < args.Length; i++)
            {
                Console.WriteLine(args[i]);
            }
        }
    }
}
```

Output:



Microsoft Visual Studio Debug Console

Mandaviya Tisha Jayesh - 25SOECE13066

Hello, World!

You entered the following 4 command line arguments:

A

B

C

D

7. Predict and write the output of the given code.

```
using System;
namespace CalculatorApplication
{
    class NumberManipulator
    {
        public void swap(ref int x, ref int y)
        {
            int temp;
```

```
        temp = x; /* save the value of x */
        x = y; /* put y into x */
        y = temp; /* put temp into y */
    }
}
class TestRef
{
    static void Main(string[] args)
    {
        NumberManipulator n = new NumberManipulator();
        /* local variable definition */
        int a = 100;
        int b = 200;
        Console.WriteLine("Before swap, value of a : {0}", a);
        Console.WriteLine("Before swap, value of b : {0}", b);
        /* calling a function to swap the values */
        n.swap(ref a, ref b);
        Console.WriteLine("After swap, value of a : {0}", a);
        Console.WriteLine("After swap, value of b : {0}", b);
        Console.ReadLine();
    }
}
}
```

Code:

```
using System;

namespace Practicals
{
    class Number
    {
        public static void Swap(ref int x, ref int y)
        {
            int temp;
            temp = x;
            x = y;
            y = temp;
        }
    }
    internal class T2_07
```

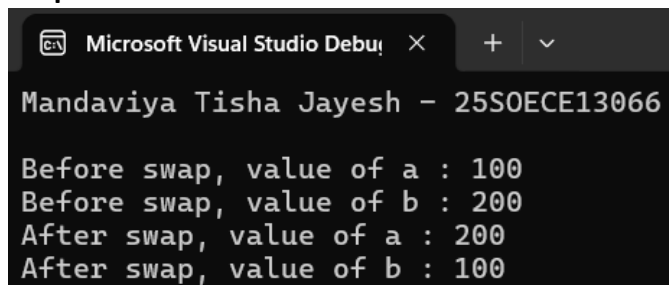
```
{
    static void Main(string[] args)
    {
        Console.WriteLine("Mandaviya Tisha Jayesh - 25SOECE13066\n");

        int a = 100;
        int b = 200;

        Console.WriteLine("Before swap, value of a : {0}", a);
        Console.WriteLine("Before swap, value of b : {0}", b);

        Number.Swap(ref a, ref b);
        Console.WriteLine("After swap, value of a : {0}", a);
        Console.WriteLine("After swap, value of b : {0}", b);
    }
}
```

Output:



Microsoft Visual Studio Debug Console output:

```
Mandaviya Tisha Jayesh - 25SOECE13066
Before swap, value of a : 100
Before swap, value of b : 200
After swap, value of a : 200
After swap, value of b : 100
```

8. Find out the error code and correct it. Write the output of the corrected code.

```
using System;
namespace CalculatorApplication
{
    class NumberManipulator
    {
        public int getValues(out int x, out int y, out int z )
        {
            Console.WriteLine("Enter the first value: ");
            x = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter the second value: ");
```

```
        y = Convert.ToInt32(Console.ReadLine());
        sum = "x" + "y" + "z";
        return "sum";
    }
}
class TestOut
{
    static void Main(string[] args)
    {
        NumberManipulator n = new NumberManipulator();
        /* local variable definition */
        int a , b, c, sum;
        /* calling a function to get the values */
        sum = n.getValues(out a, out b, out c);
        Console.WriteLine("After method call, value of a : {0}", a);
        Console.WriteLine("After method call, value of b : {0}", b);
        Console.WriteLine("After method call, value of c : {0}", c);
        Console.WriteLine("Sum : {0}", );
    }
}
}
```

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T2_08
```

```
    {
```

```
        class NumberManipulator
```

```
        {
```

```
            public int getValues(out int x, out int y, out int z)
```

```
            {
```

```
                Console.Write("Enter the first value: ");
```

```
                x = Convert.ToInt32(Console.ReadLine());
```

```
                Console.Write("Enter the second value: ");
```

```
                y = Convert.ToInt32(Console.ReadLine());
```

```
                Console.Write("Enter the third value: ");
```

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```
        z = Convert.ToInt32(Console.ReadLine());

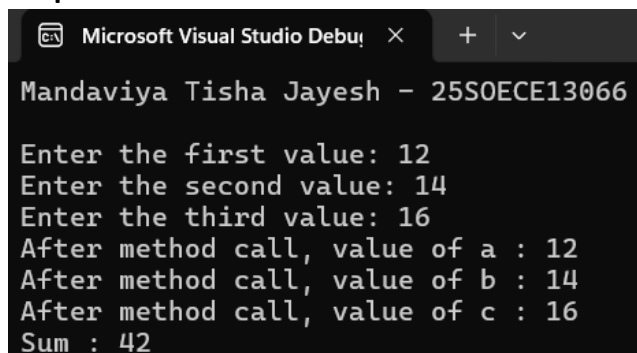
        int sum = x + y + z;
        return sum;
    }
}

class TestOut
{
    static void Main(string[] args)
    {
        Console.WriteLine("Dafda Priyanka Tarunbhai - 25SOECE13051\n");
        NumberManipulator n = new NumberManipulator();
        int a, b, c, sum;

        sum = n.getValues(out a, out b, out c);

        Console.WriteLine("\nAfter method call, value of a : {0}", a);
        Console.WriteLine("After method call, value of b : {0}", b);
        Console.WriteLine("After method call, value of c : {0}", c);
        Console.WriteLine("Sum : {0}", sum);
    }
}
}
```

Output:



Microsoft Visual Studio Debug Console window showing the output of the program. The title bar reads "Mandaviya Tisha Jayesh - 25SOECE13066". The output text is as follows:

```
Enter the first value: 12
Enter the second value: 14
Enter the third value: 16
After method call, value of a : 12
After method call, value of b : 14
After method call, value of c : 16
Sum : 42
```

9. Given an array A containing $2*N+2$ positive numbers, out of which $2*N$ numbers exist in pairs whereas the other two number occur exactly once and are distinct. Find the other two numbers.

Example 1:Input:

N = 2

arr[] = {1, 2, 3, 2, 1, 4}

Output:

3 4

Explanation:

3 and 4 occur exactly once.

Example 2:

Input:

N = 1

arr[] = {2, 1, 3, 2}

Output:

1 3

Explanation:

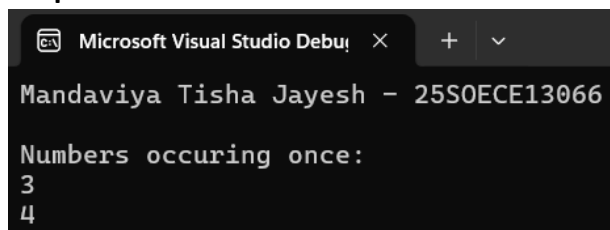
1 3 occur exactly once.

Code:

```
using System;
namespace Practicals
{
    internal class T2_09
    {
        public static void T2_09Main()
        {
            int[] arr = { 1, 2, 3, 2, 1, 4 };
            Console.WriteLine("Numbers occuring once:");
            for(int i=0; i < arr.Length; i++)
            {
                int count= 0;
                for (int j=0; j < arr.Length; j++)
                {
                    if (arr[i] == arr[j])
                    {
                        count++;
                    }
                }
            }
        }
    }
}
```

```
        if (count == 1)
        {
            Console.WriteLine(arr[i] + " ");
        }
    }
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
Numbers occuring once:
3
4
```

10. Given a matrix `mat[][]` of size `N x M`, where every row and column is sorted in increasing order, and a number `X` is given. The task is to find whether element `X` is present in the matrix or not.

Example 1:

Input:

`N = 3, M = 3`

`mat[][] = 3 30 38`

`44 52 54`

`57 60 69`

`X = 62`

Output:

0

Explanation:

62 is not present in the matrix, so output is 0

Example 2:

Input:

`N = 1, M = 6`

`mat[][] = 18 21 27 38 55 67`

`X = 55`

Output:

1

Explanation:

55 is present in the

matrix at 5th cell.

Your Task:

You don't need to read input or print anything. You just have to complete the function `matSearch()` which takes a 2D matrix `mat[][]`, its dimensions `N` and `M` and integer `X` as inputs and returns 1 if the element `X` is present in the matrix and 0 otherwise.

Expected Time Complexity: $O(N+M)$.

Expected Auxiliary Space: $O(1)$.

Constraints:

$1 \leq N, M \leq 1005$

$1 \leq \text{mat}[][] \leq 10000000$

$1 \leq X \leq 10000000$

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T2_10
```

```
    {
```

```
        static int MatSearch(int[,] mat, int n, int m, int x)
```

```
        {
```

```
            int row = 0;
```

```
            int col = m - 1;
```

```
            while(row < n && col >= 0)
```

```
            {
```

```
                if (mat[row, col] == x)
```

```
                {
```

```
                    return 1;
```

```
                }
```

```
                else if (mat[row, col] == x)
```

```
                {
```

```
                    col--;
```

```
                }
```

```
                else
```

```
                {
```

```
                    row++;
```

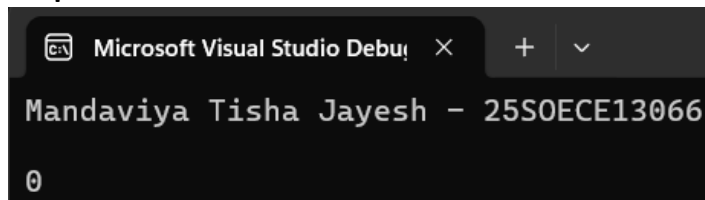
```
                }
```

```
            }
```



```
        return 0;
    }
    public static void T2_10Main()
    {
        int[,] mat =
        {
            {3,30,38},
            {44,52,54},
            {57,60,65}
        };
        int result = MatSearch(mat, 3, 3, 50);
        Console.WriteLine(result);
    }
}
```

Output:



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0

11. Write a program to find the sum of N elements of an Array.

Code:

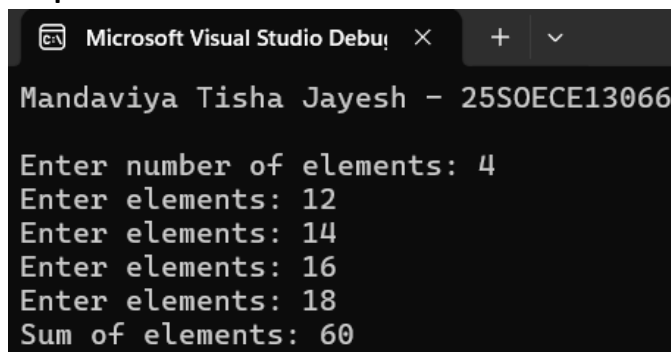
```
using System;

namespace Practicals
{
    internal class T2_11
    {
        public static void SumOfArray()
        {
            int n, sum = 0;
            Console.Write("Enter number of elements: ");
            n = Convert.ToInt32(Console.ReadLine());
            int[] arr = new int[n];

            for (int i = 0; i < n; i++)
            {
                Console.Write("Enter elements: ");
            }
        }
    }
}
```

```
        arr[i] = Convert.ToInt32(Console.ReadLine());
        sum += arr[i];
    }
    Console.WriteLine("Sum of elements: {0}", sum);
}
}
```

Output:



```
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Enter number of elements: 4
Enter elements: 12
Enter elements: 14
Enter elements: 16
Enter elements: 18
Sum of elements: 60
```

12. Write a program to find the element from an Array and print 1 if element is found else print 0.

Code:

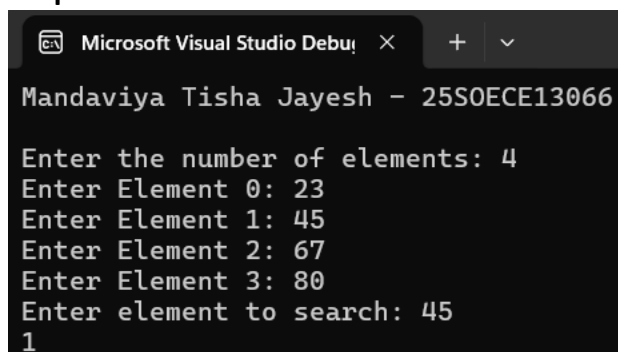
```
using System;

namespace Practicals
{
    internal class T2_12
    {
        public static void T2_12Main()
        {
            int n, search;
            int found = 0;
            Console.Write("Enter total number of Elements:");
            n = Convert.ToInt32(Console.ReadLine());

            int[] arr = new int[n];
            for (int i = 0; i < n; i++)
            {
                Console.Write($"Enter Element{i}:");
                arr[i] = Convert.ToInt32(Console.ReadLine());
            }
        }
    }
}
```

```
Console.WriteLine("Enter an element to search:");
search = Convert.ToInt32(Console.ReadLine());
for (int i = 0; i < n; i++)
{
    if (arr[i] == search)
    {
        found = 1;
        break;
    }
}
Console.WriteLine(found);
}
```

Output:



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Enter the number of elements: 4
Enter Element 0: 23
Enter Element 1: 45
Enter Element 2: 67
Enter Element 3: 80
Enter element to search: 45
1

13. Write a Program that will accept the amount and find how many minimum no of notes you required for that.

(Using the rupee notes of 1, 2, 5, 10, 20, 50, 100, 200, 500, 2000)

Input: 5748

Output:

Notes of Rs.2000 = 2 Notes of Rs.500 = 3 Notes of Rs.200 = 1

Notes of Rs.20 = 2 Notes of Rs.10 = 0 Notes of Rs.5 = 1

Notes of Rs.2 = 1 Notes of Rs.1 = 1

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

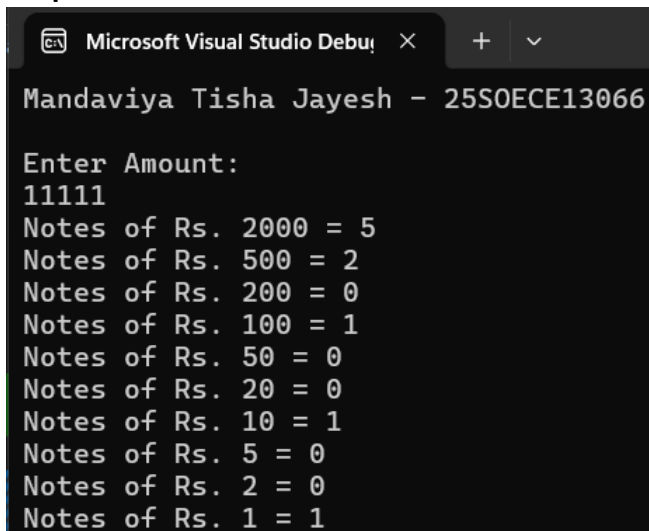
```
    internal class T2_13
```

```
{
    public static void T2_13Main()
    {
        Console.WriteLine("Enter Amount:");
        int amount=Convert.ToInt32(Console.ReadLine());

        int[] notes = { 2000, 500, 200, 100, 50, 20, 10, 5, 2, 1};

        foreach(int note in notes)
        {
            int count = amount / note;
            Console.WriteLine($"Notes of Rs. {note} = {count}");
            amount = amount % note;
        }
    }
}
```

Output:



```
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Enter Amount:
11111
Notes of Rs. 2000 = 5
Notes of Rs. 500 = 2
Notes of Rs. 200 = 0
Notes of Rs. 100 = 1
Notes of Rs. 50 = 0
Notes of Rs. 20 = 0
Notes of Rs. 10 = 1
Notes of Rs. 5 = 0
Notes of Rs. 2 = 0
Notes of Rs. 1 = 1
```

14. Write a Program to find the eligibility of admission for a professional course based on the following criteria:
- Marks in Maths ≥ 65
 - Marks in Phy ≥ 55
 - Marks in Chem ≥ 50 and
 - Total in all three subject ≥ 180 or
 - Total in Math and Physics ≥ 140

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INPUT:

Input the marks obtained in Maths :72

Input the marks obtained in Physics :65

Input the marks obtained in Chemistry :51

OUTPUT:

The candidate is eligible for admission.

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T2_14
```

```
    {
```

```
        public static void T2_14Main()
```

```
        {
```

```
            Console.WriteLine("Input the marks obtained in Maths:");
```

```
            int maths = Convert.ToInt32(Console.ReadLine());
```

```
            Console.WriteLine("Input the marks obtained in Physics:");
```

```
            int physics = Convert.ToInt32(Console.ReadLine());
```

```
            Console.WriteLine("Input the marks obtained in Chemistry:");
```

```
            int chemistry = Convert.ToInt32(Console.ReadLine());
```

```
            int total= maths + physics + chemistry;
```

```
            int mp = maths + physics;
```

```
            if((maths>=65 && physics>=55 && chemistry >= 50 && total>=180) || (maths>=65 && physics>=55 && mp>=140))
```

```
            {
```

```
                Console.WriteLine("The candidate is eligible for admission.");
```

```
            }
```

```
            else
```

```
            {
```

```
                Console.WriteLine("The candidate is NOT eligible for admission.");
```

```
            }
```

```
        }
```

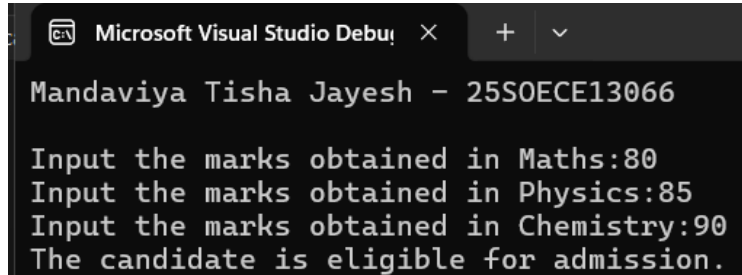
```
    }
```

```
}
```

25SOECE13066

Enterprise Computing Through .NET Framework (CE525)

Output:



```
Microsoft Visual Studio Debug Console
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Input the marks obtained in Maths:80
Input the marks obtained in Physics:85
Input the marks obtained in Chemistry:90
The candidate is eligible for admission.
```

15. Write a Program which accepts name from the user and prints the same

INPUT : R K University

OUTPUT: R K University

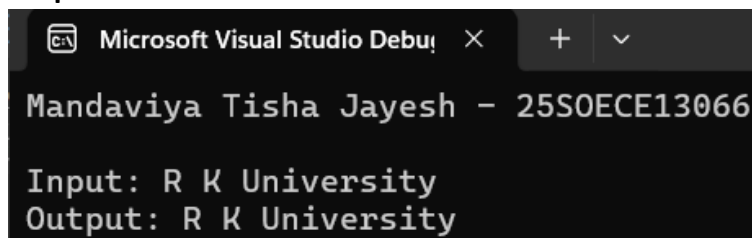
Code:

```
using System;
```

```
namespace Practicals
```

```
{
    internal class T2_15
    {
        public static void T2_15Main()
        {
            Console.Write("Input: ");
            string name = Console.ReadLine();
            Console.WriteLine("Output: " + name);
        }
    }
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066

Input: R K University
Output: R K University
```